

ODRE PQC

公開技術ホワイトペーパー

ODRE PQC 0.3.1 RC1 · Release 63

検証範囲、Authority、信頼境界、対応OS、FULL AUM、実測性能をまとめます。

2026年9月 · Release 63 技術基準

1. 文書基準と最終判定

Release 63 は Final Customer Delivery の Authority です。稼働中 Production のポインターは Release 60 であり、両者を混同しません。

Authority	Verified state
Final Customer Delivery	Release 63 · g269 · 6/6 OS PASS
Active Production	Current 60 · Previous 59 · COMMITTED · Health PASS
Policy	Generation 1080 · Consumer Pin SYNCED
FULL AUM	g270 · A-R isolated PASS · destructive Production test NOT RUN BY POLICY

- Customer Delivery 技術判定は YES、公式対応OSは6/6 PASSです。
- Negative Gate 18/18 PASS。Release 63、Sealed Core、g269 Evidence は確定後に変更されていません。
- FULL AUM A-R はインストール済み updater byte と隔離 fixture で検証し、Production 破壊試験はポリシーにより未実施です。

Engineering interpretation

A later PASS does not erase an earlier FAIL. Every statement is limited to the artifact, platform, environment and gate identified by its evidence.

2. 製品の目的

ODRE PQC は、PQC保護実行を署名済みartifact、runtime identity、policy generation、consumer pin、License/Lease/Unit Authority に結合します。前提が失敗しても安全でない fallback は行いません。

- Data Plane: Commercial Native と承認済み Sealed Core が保護演算を実行します。
- Control Plane: policy、pin、identity、商用権限が実行可否を決定します。
- Evidence Plane: manifest、receipt、journal、canonical status が PASS/FAIL/NOT RUN の根拠を保持します。

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3. セキュリティアーキテクチャ

Gateway、Native Runtime、Sealed Core、Policy、License、AUM は分離されますが、protected request 時には同一 Authority に結合されます。

- 必須artifactの欠落・改変は install/apply 前に失敗します。
- policy と consumer pin の不一致では実行0、partial apply 0です。
- Native/Core readiness または protected request の失敗時は COMMIT を禁止します。
- Sealed Core は変更禁止ですが、package/install/load/test の対象です。

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4. Protected Request 契約

認証済み要求は、権限、identity、policy同期、Native readiness、Core load を確認した後のみ実行されます。

- 成功した protected request ごとに business handler は exactly once で実行されます。
- unauthorized execution、plaintext fallback、暗黙の classical downgrade は0です。
- health code だけでは不十分で、protected request が commit gate に含まれます。

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5. Release パッケージ信頼チェーン

選択基準は署名済み Authority と正確な SHA-256 であり、ファイル名や更新時刻ではありません。

Generation	Role
60	Active Production baseline
61	Ubuntu runtime-manifest pin repair lineage
62	Immutable intermediate successor
63	Final Customer Delivery authority

- archive → inventory/manifest → envelope/CMS → installer binding → installed load → protected integration → final ZIP seal。
- Release 60/61/62 は上書きせず、Release 63 を単調増加 successor として作成しました。
- Installer Execution、Installation Integration、Runtime Integration、Customer Delivery は別々のGateです。

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6. FULL AUM ライフサイクル

DISCOVER、VERIFY、STAGE、SWITCH、HEALTH、COMMIT、ROLLBACK、RECOVERY が durable transaction を構成します。

- single writer lock が同時 apply を防止します。
- コピー途中の candidate は VERIFIED になりません。
- atomic status と durable journal が partial JSON や偽 COMMITTED を防止します。
- staging 前に容量を確認し、rollback asset を削除しません。

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7. サービスと管理

Windows サービスは SCM lifecycle を使用し、start_server は exe/python の直接起動で契約を迂回しません。

- Running service は再起動せず、Stopped のみ SCM で起動します。
- Admin Update は認証済み・dynamic・read-only です。
- Production: current 60、previous 59、floor 60、policy 1080、pin SYNCED、COMMITTED/PASS、health 8000/18443 HTTP 200。

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8. 公式対応プラットフォーム

同一の package/authority/protected-request 契約が Windows Server 2022、Ubuntu 22.04/24.04、Debian 12、Rocky Linux 9.8、AlmaLinux 9.8 で PASS しました。

OS	Result	Receipt SHA-256
Windows Server 2022	PASS	5E390E2062034FDE03BFA304D6A64EEAA191F79CB292E0EA913086CF5CB30489
Ubuntu 22.04	PASS	2E19C70F2992A1FA520E757DFC64A2FDB78DDEDBAB4E6AA4F943954D7B6E627D
Ubuntu 24.04	PASS	4AFBB78E6AB65D7D9389FA2B337C7E7350828AD69D827011CA399BAB266BC1B4
Debian 12	PASS	2597605241A07F53282EE0CEE6E5EF101919E178AAD00CBC06FF7FEBB2060293
Rocky Linux 9.8	PASS	701244BC819DA109AA190621F83EC91C36C5EDC0129041D80509F9AA952070AE
AlmaLinux 9.8	PASS	EBDBA21B870BD33032C26420669FC3F313890F45548B3369EB974C9E8FEB8563

- SCM/ACL または service/permission adapter は trust model を変更しません。
- Container、Kubernetes、その他の Windows/Linux version はこの判定範囲外です。

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9. Fail-Closed と Negative Test

ZIP/SHA改変、CMS失敗、inventory不一致、policy drift、duplicate/lower generation、runtime失敗、容量不足は安全側で停止します。

- REJECTED_PRE_APPLY の Production change は0です。
- switch後 health失敗では以前の Product/policy/pin/runtime を復元します。
- 初回FAIL Evidence は修正・再試験後も保持します。

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10. Evidence と解釈範囲

g269 は Final Customer Delivery、g270 は限定された FULL AUM と live read-only Admin を証明します。

- Updater 208/208、drop-in 14/14、12-A/Admin 19/19 PASS。
- Drive upload/readback と Notion 2ページ append/readback PASS。Drive再download SHAはポリシーにより NOT RUN。
- FIPS 140-3、Common Criteria、普遍的性能・全プラットフォーム対応は主張しません。

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実測 RPS / レイテンシ

g108 の Windows Server 2022 loopback candidate 試験です。Final Release 63 で再測定した値ではなく、WANや顧客hardwareへ一般化しません。

Test	Measured value
Environment	Windows Server 2022 build 20348 · AMD EPYC 9454P · 4 cores · 16 GiB · loopback
Protected 1W max stable mean	1,176.9337 RPS (range 1,163.8582–1,190.0091)
Baseline 1W max stable mean	3,442.9391 RPS (range 3,413.4320–3,472.4462)
Same concurrency 2	Baseline 3,442.9391 / Protected 1,104.7263 RPS · reduction 67.9133%
Latency c2 baseline → protected	p50 0.5468→1.7223 ms · p95 0.7238→2.2881 ms · p99 0.9382→2.6597 ms
Protected security	200,128 success / 200,128 handler · invariant failure 0 · plaintext HTTP 426 · fallback NONE
Protected 2W / 4W	NOT_MEASURED · single_worker_lease_unavailable fail-closed
Baseline scaling reference	2W 5,035.1443 RPS c8 · 4W 5,475.0515 RPS c32 · zero error

- Same business_handler, POST /secure/echo and serialization were used for Baseline and Protected.
- The 2W/4W protected values were not estimated or multiplied.
- These figures do not support a claim that PQC overhead is negligible.

Conclusion

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Release 63 Final Customer Delivery is technically PASS. Active Production Release 60 remains a separate committed authority. Unexecuted scopes remain explicitly NOT RUN.